

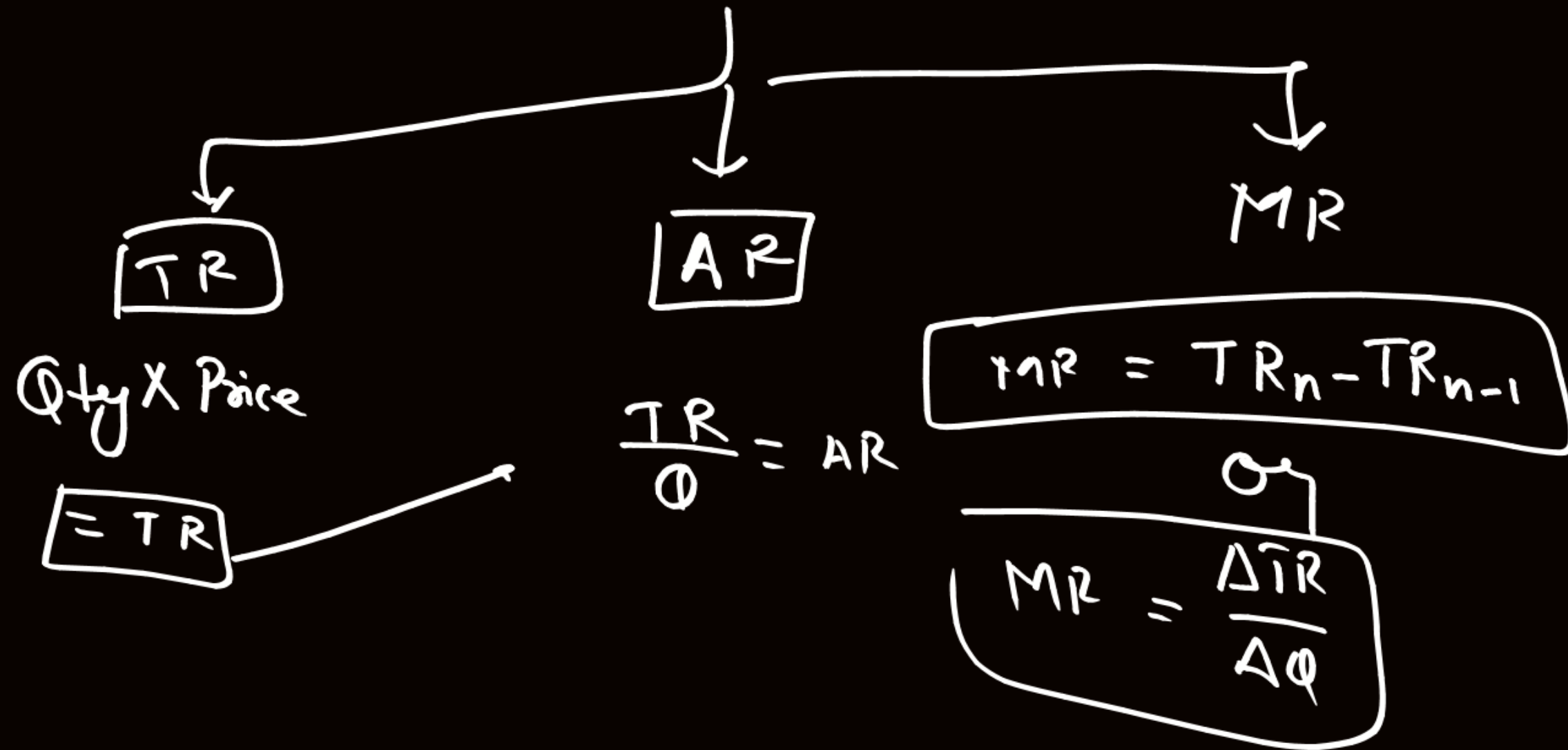


MICROECONOMICS

REVENUE

Chapter 7 | Lecture 3

Revenue



Price = AR = MR

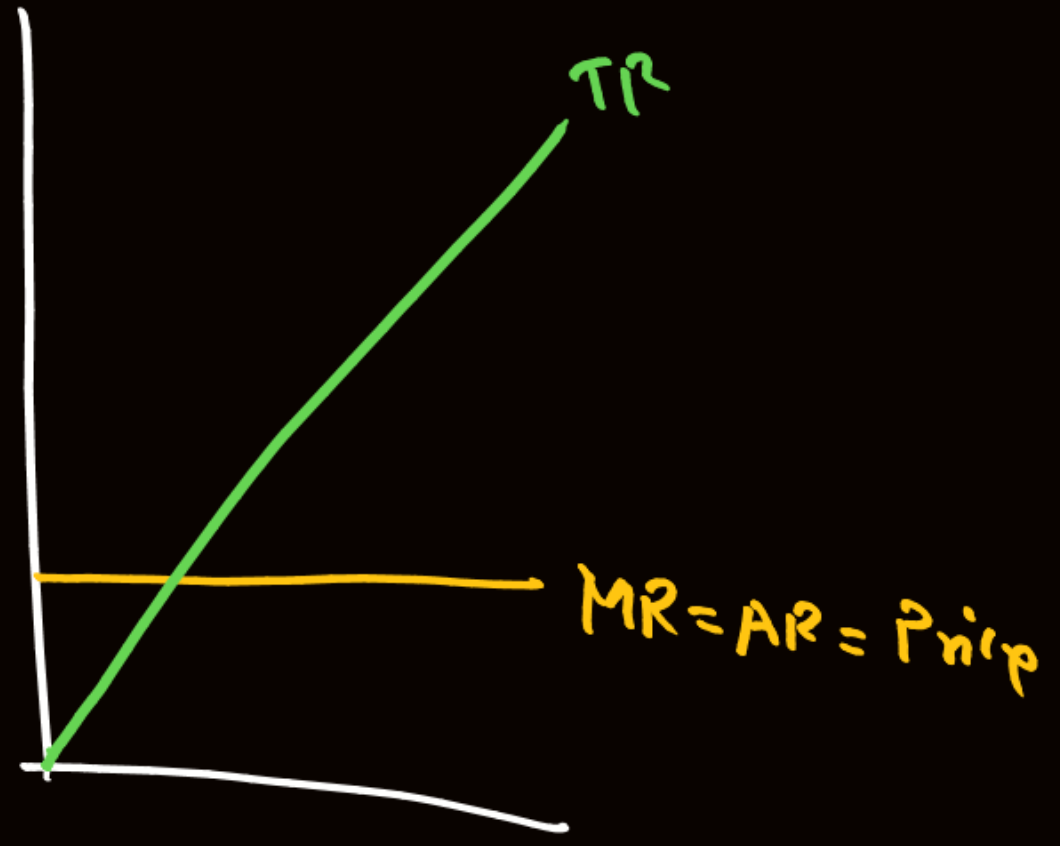
x

Perfect Competition

Imperfect Competition

Sales ↑
Price ↓

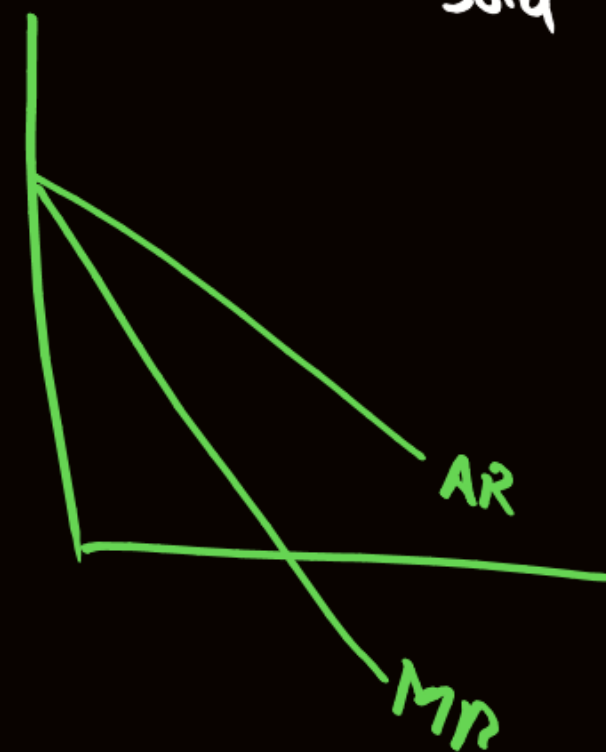
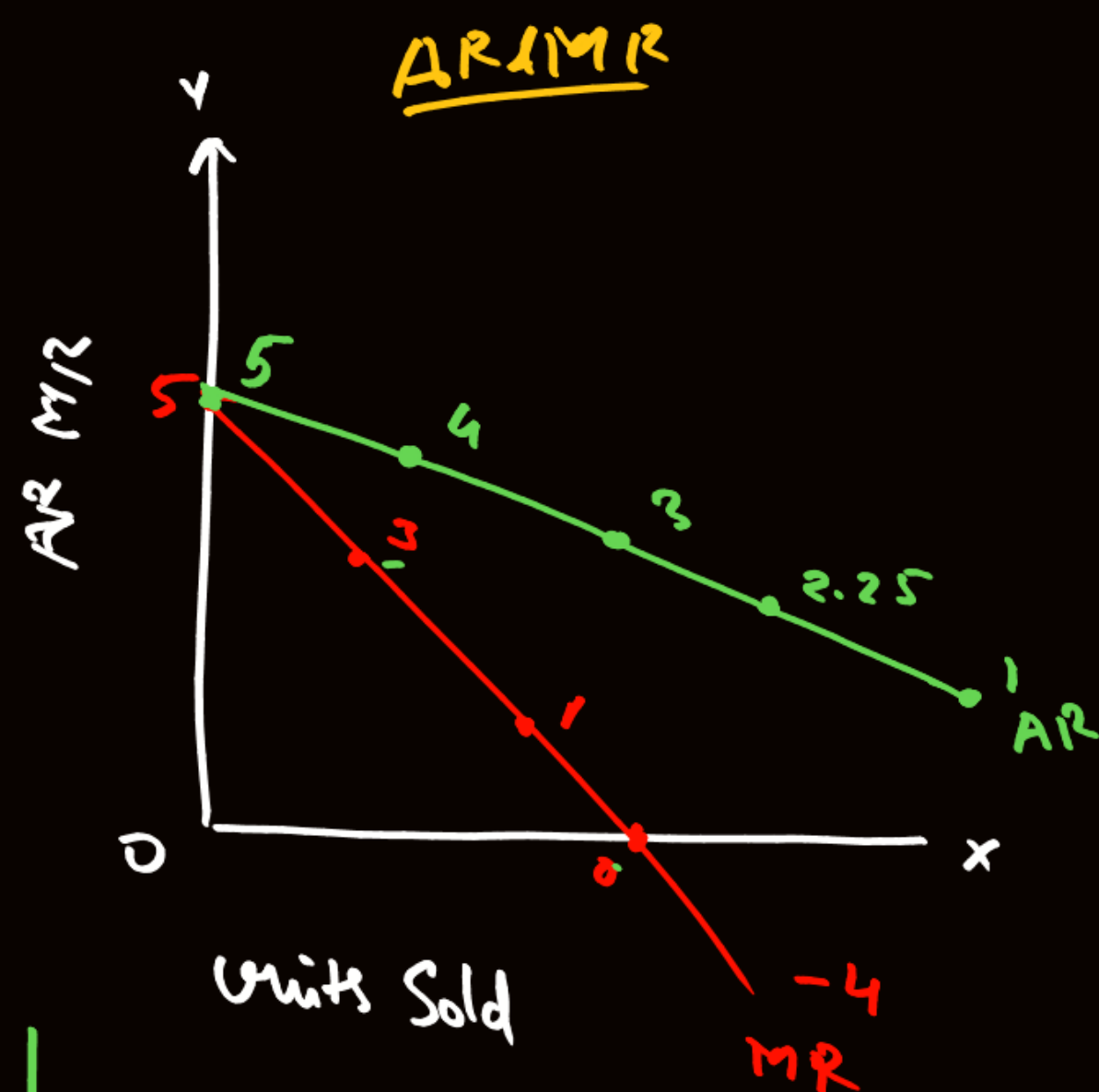
Unit	Price/AR	TR	MR
1	5	5	5
2	5	10	5
3	5	15	5
4	5	20	5
5	5	25	5



Units	Price
1	10
2	9

Units	Price/AR	TR	MR
1	5	5	5
2	4	8	3
3	3	9	1
4	2.25	9	0
5	1	5	-4

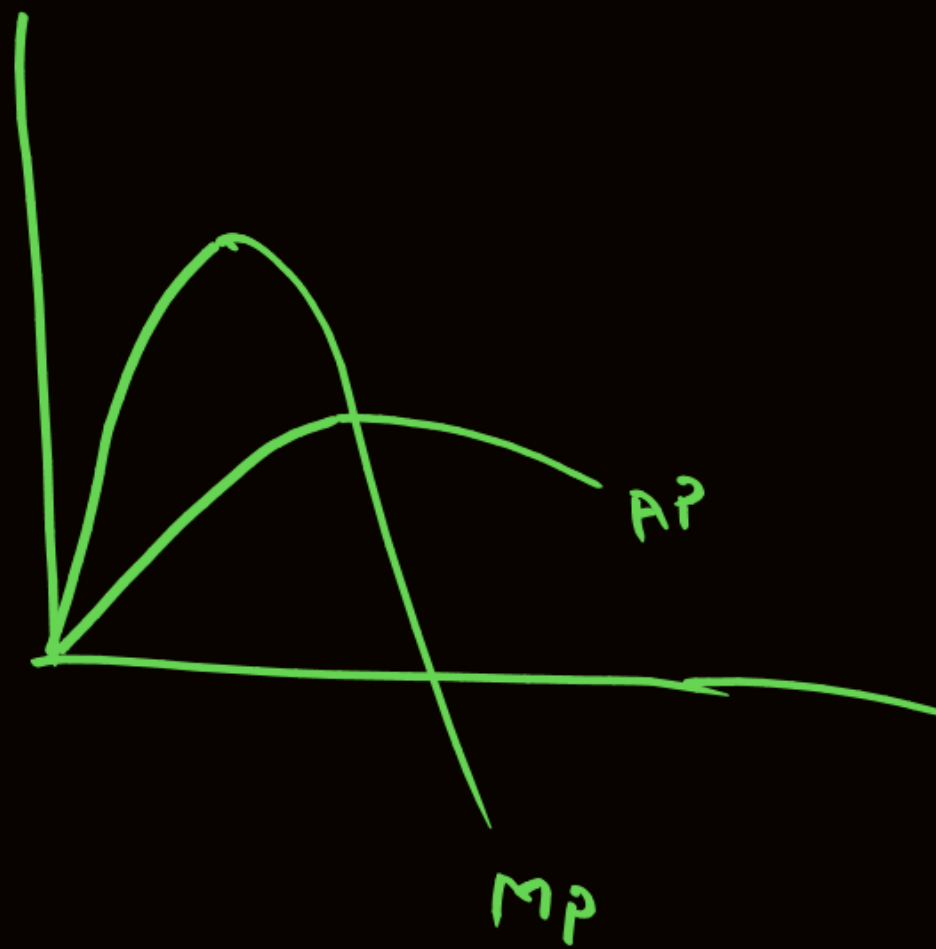
↓ $MR < AR$



$$1.) \quad \cancel{4} \rightarrow 25 \rightarrow \textcircled{1} = \textcircled{25} \checkmark$$

$$x \quad 2.) \quad \begin{array}{c} \cancel{4} \cancel{0} \cancel{4} \\ \cancel{0} \cancel{0} \end{array} \rightarrow 21 \rightarrow \textcircled{5} = \textcircled{25} \checkmark$$

$\textcircled{\text{MR}} \text{L}$

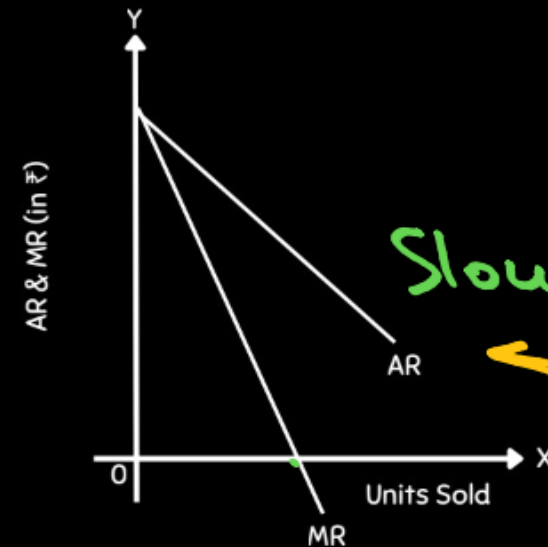


Relationship Between AR & MR (When Price Falls with rise in output)

AR & MR (When price falls with rise in output)

Units Sold	Price / AR (₹)	TR (₹)	MR (₹)
1	5	5	5
2	4	8	3
3	3	9	1
4	2.25	9	0
5	1	5	-4

Relationship Between AR & MR
AR and MR (When Price Falls with rise in output)



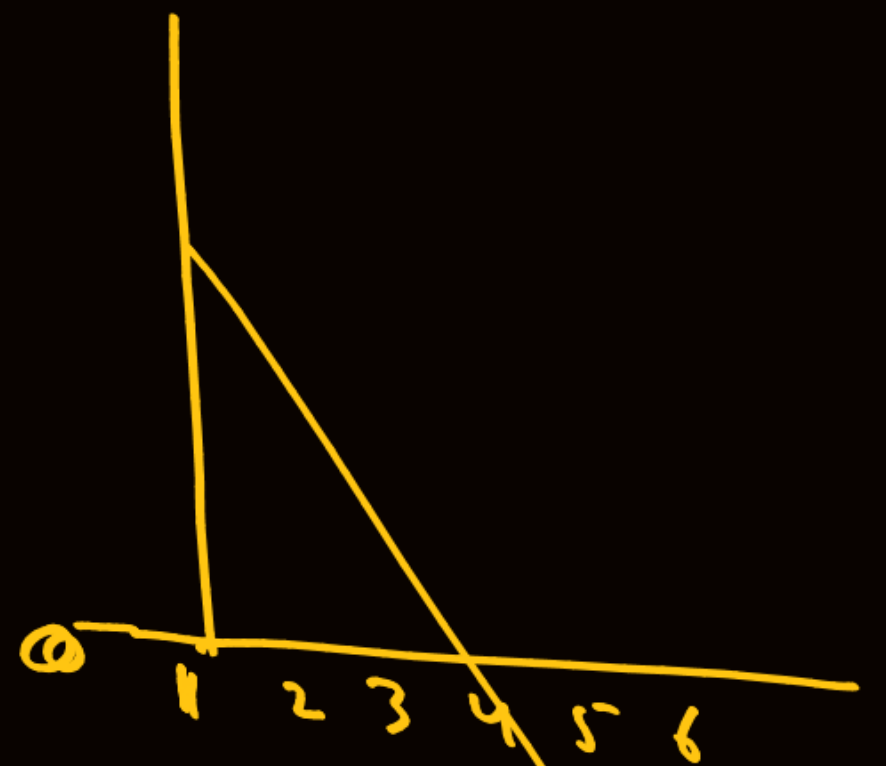
Price ↓ Qty ↑



Relationship Between TR & MR (When Price remains Constant)

- ✍ Downward-Sloping AR Curve: In imperfect competition, the AR curve (demand curve) slopes downward, meaning the firm must lower the price to sell additional units. $P \downarrow$ $Q \uparrow$
- ✍ MR < AR: Marginal Revenue (MR) is always less than Average Revenue (AR) because lowering the price to sell more units reduces the revenue gained from previous units.
- MR Declines Faster than AR: As output increases, MR declines at a faster rate than AR due to the need to reduce price for all units sold.
- MR Can Become Negative: MR can turn negative if the loss in revenue from reducing the price exceeds the gain from selling additional units.
- ✍ Profit Maximization: A firm maximizes profit by producing where MR = MC, even though MR < AR. Next Chapter
- AR Represents Price: AR is equal to the price at which units are sold, but MR reflects the incremental change in TR from selling one more unit.

$$\frac{100}{100} \times$$



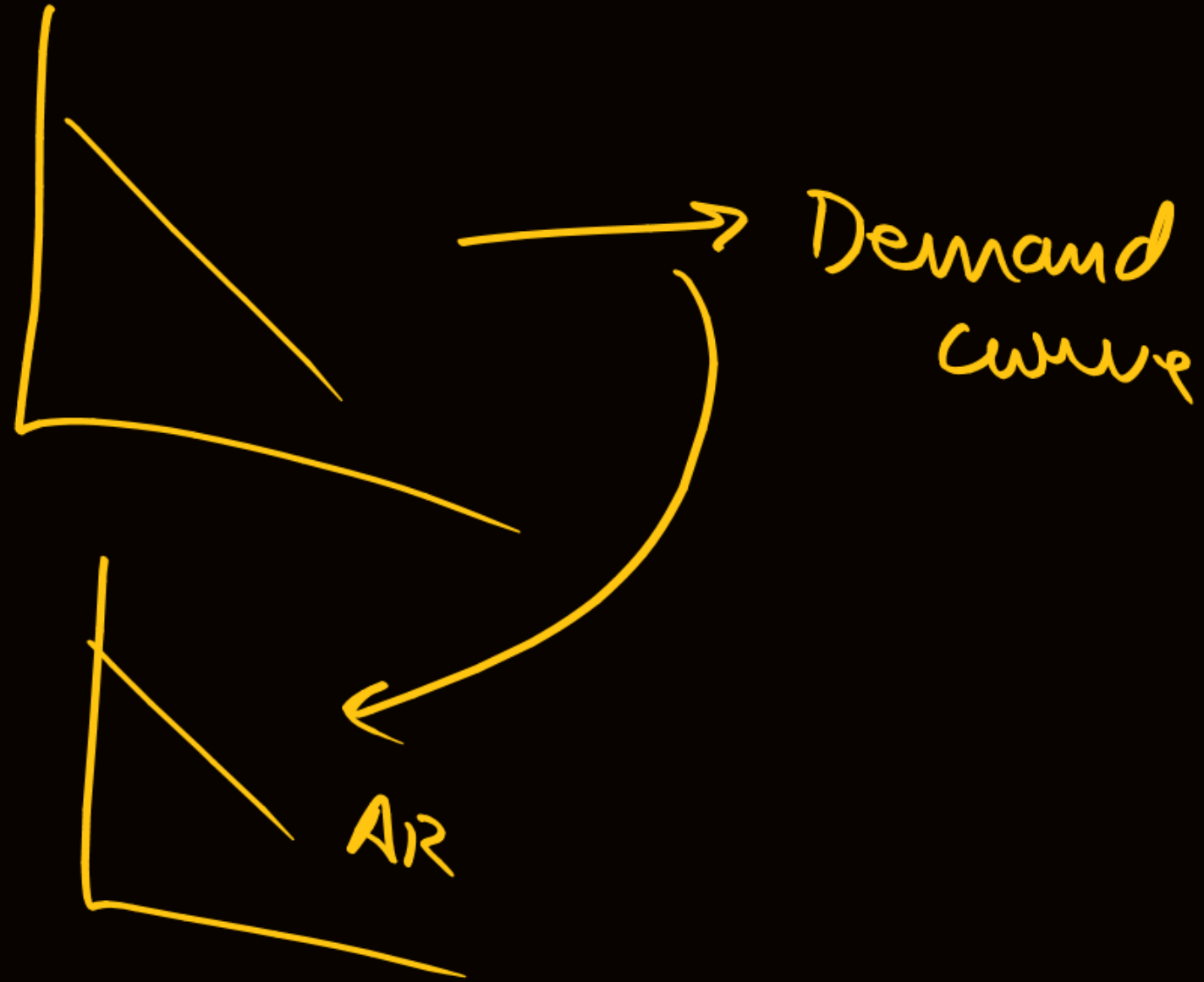
-4 MV



MV

$\uparrow \downarrow \rightarrow$ Inverse Relation

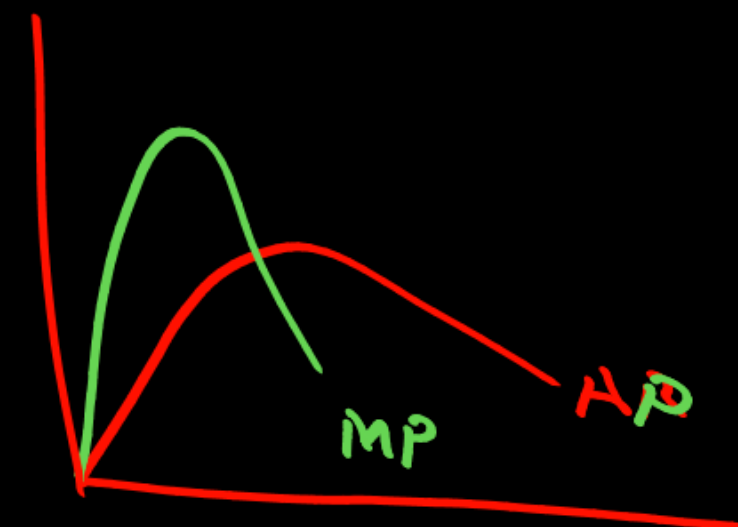
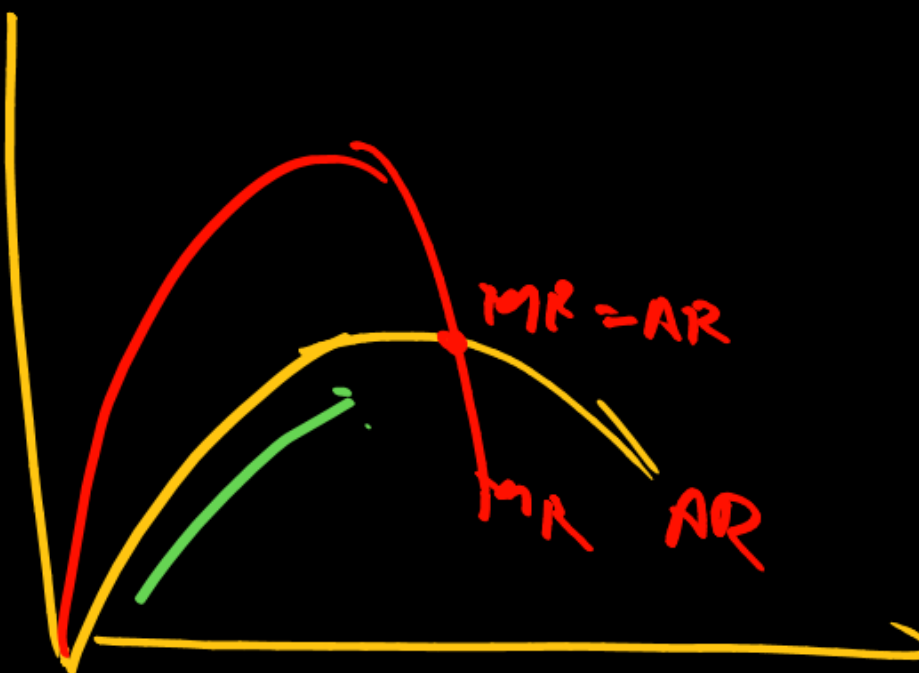
$Q \uparrow P \downarrow$

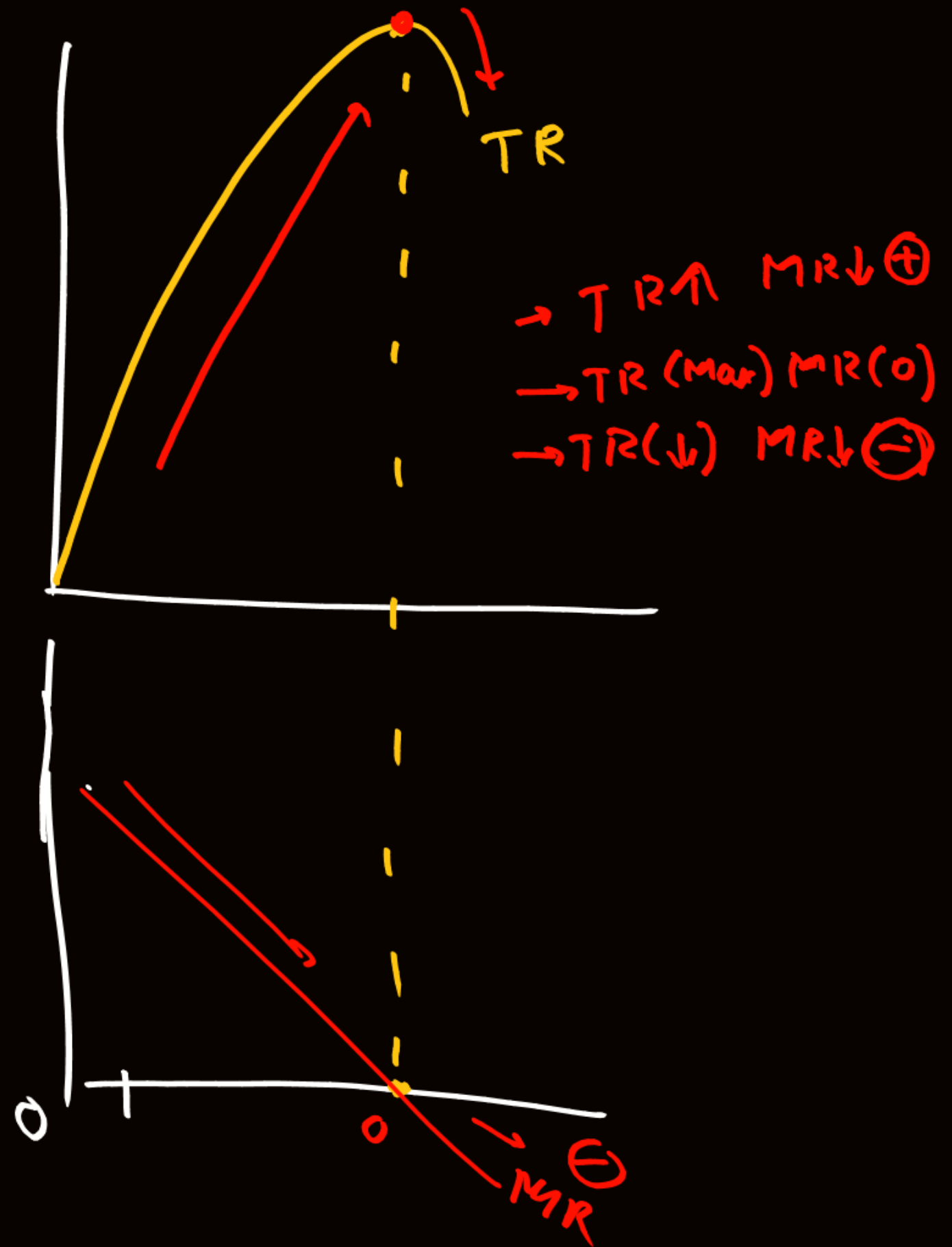
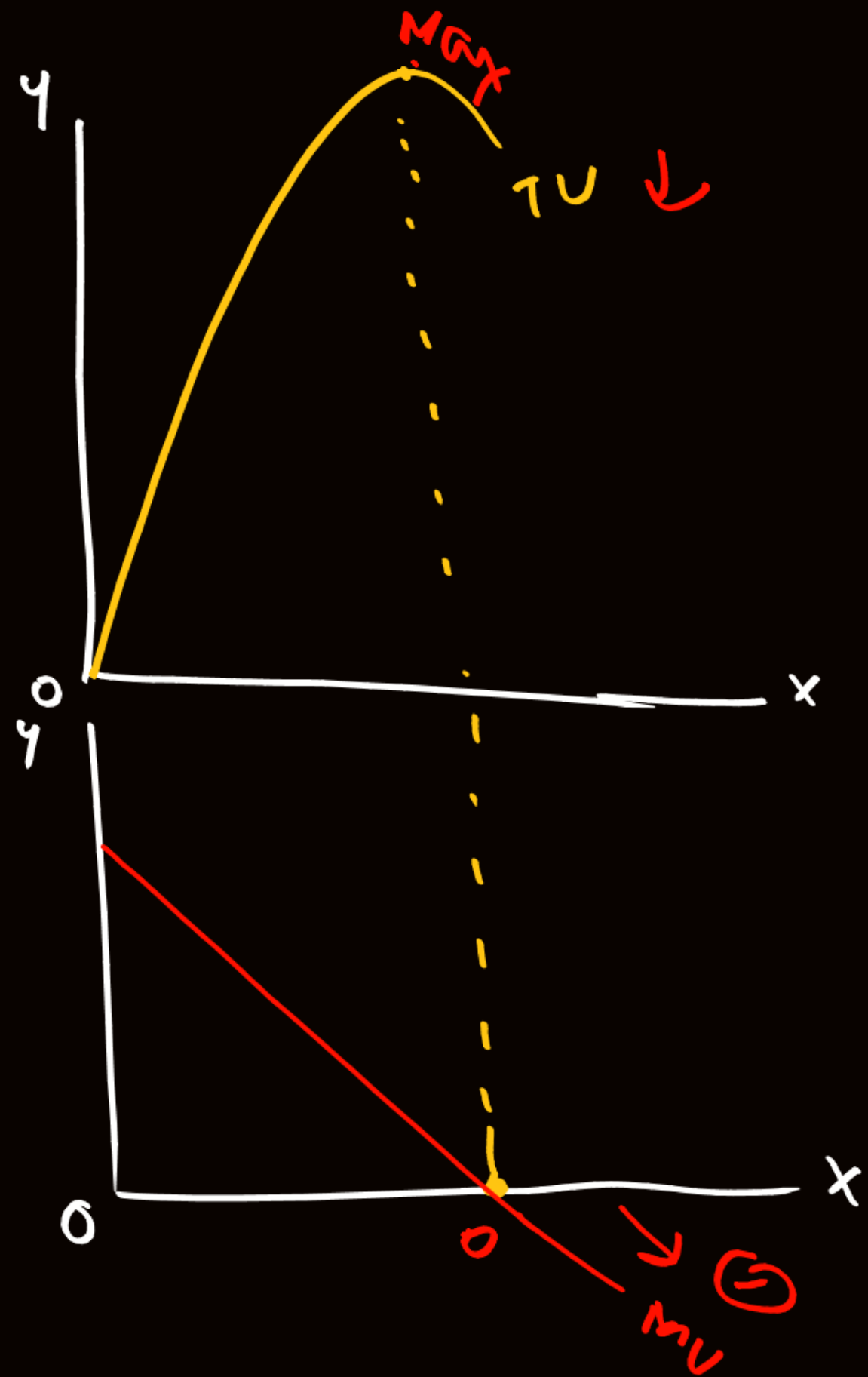


General Relationship between AR & MR

WHEN NOTHING IS MENTIONED ABOUT THE NATURE OF PRICE IS IT FALLING OR SAME.

- AR increases as long as MR is higher than AR (or when $MR > AR$ increases).
- AR is maximum and constant when MR is equal to AR (or when $MR = AR$, AR is maximum).
- AR falls when MR is less than AR (or when $MR < AR$, AR falls).



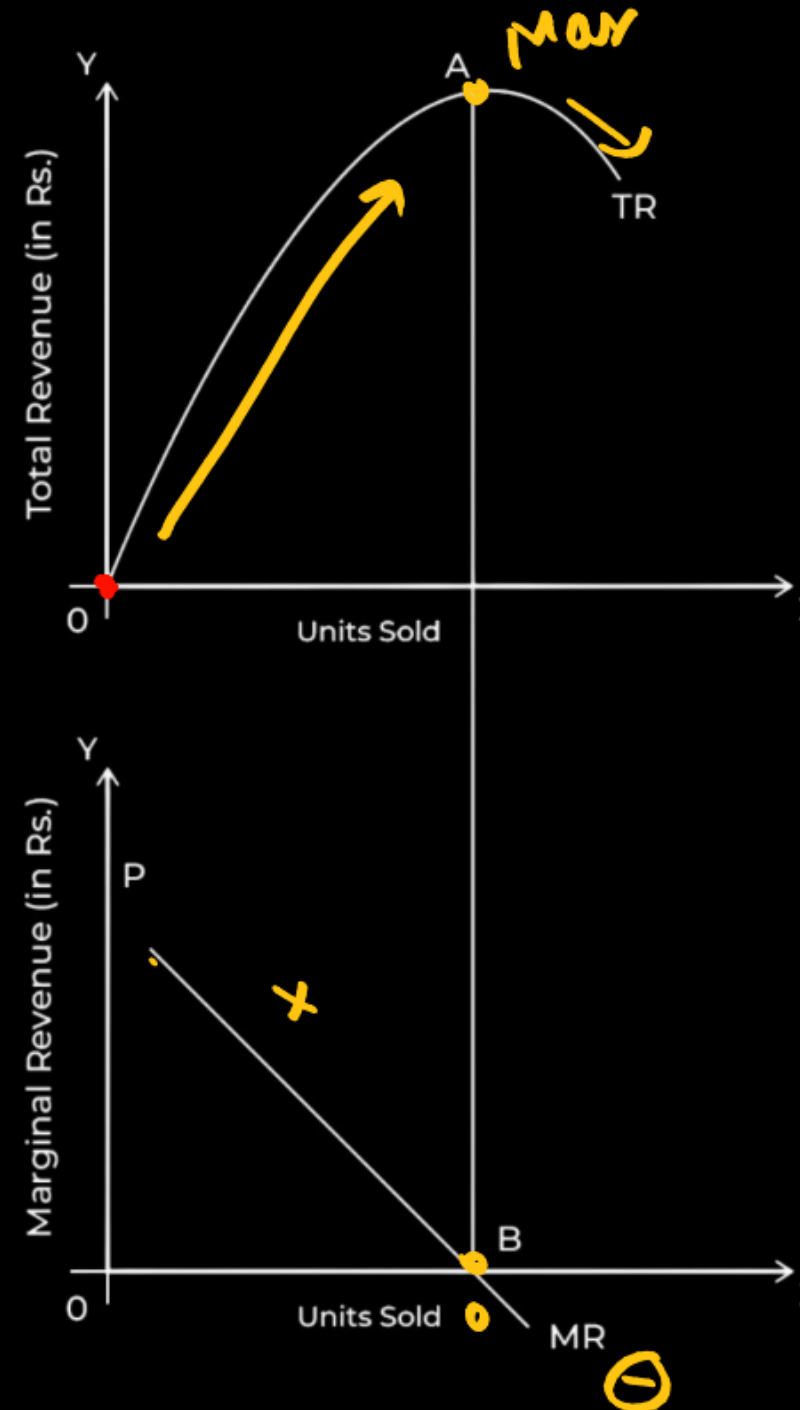


Relationship Between TR & MR (When Price Falls with rise in output)

TR and MR (When Price Falls with rise in output)

Units Sold	Price / AR (₹)	TR (₹)	MR (₹)
1	5	5	5
2	4	8	3
3	3	9	1
4	2.25	9	0
5	1	5	-4

TR $\uparrow$ $\rightarrow$ 5 $\leftarrow$ MR



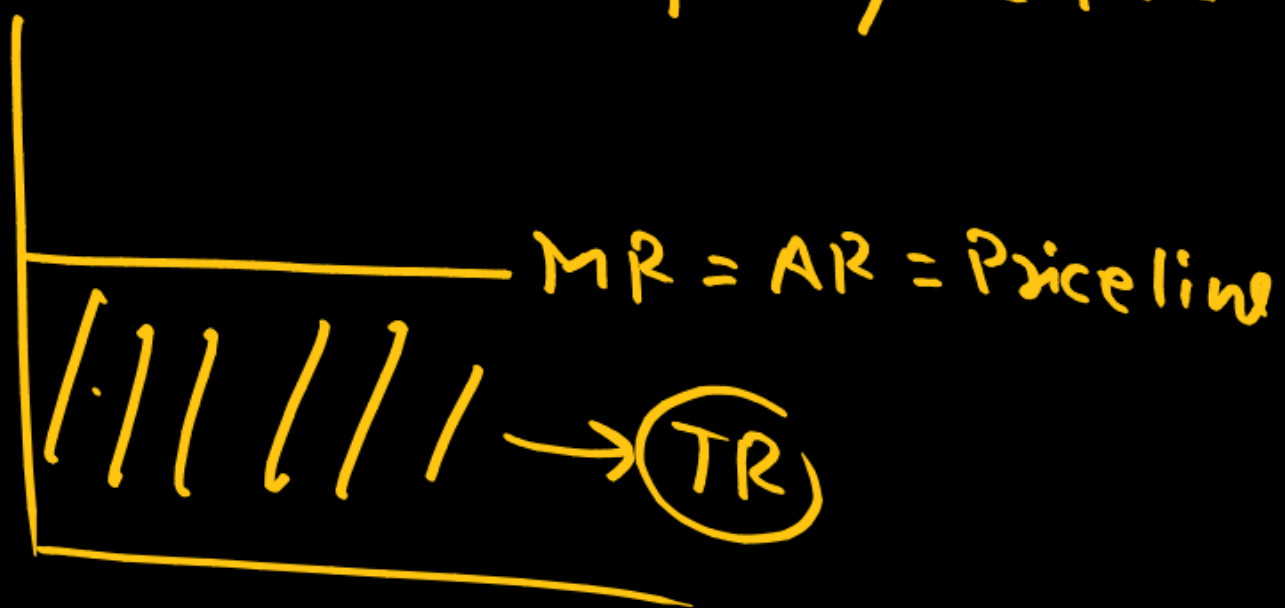
The relationship can be summed up as under:

- As long as MR is positive, TR increases (or when TR rises, MR is positive).
- When MR is zero, TR is at its maximum point (or when TR is maximum, MR is zero).
- When MR becomes negative, TR starts falling (or when TR falls, MR is negative).

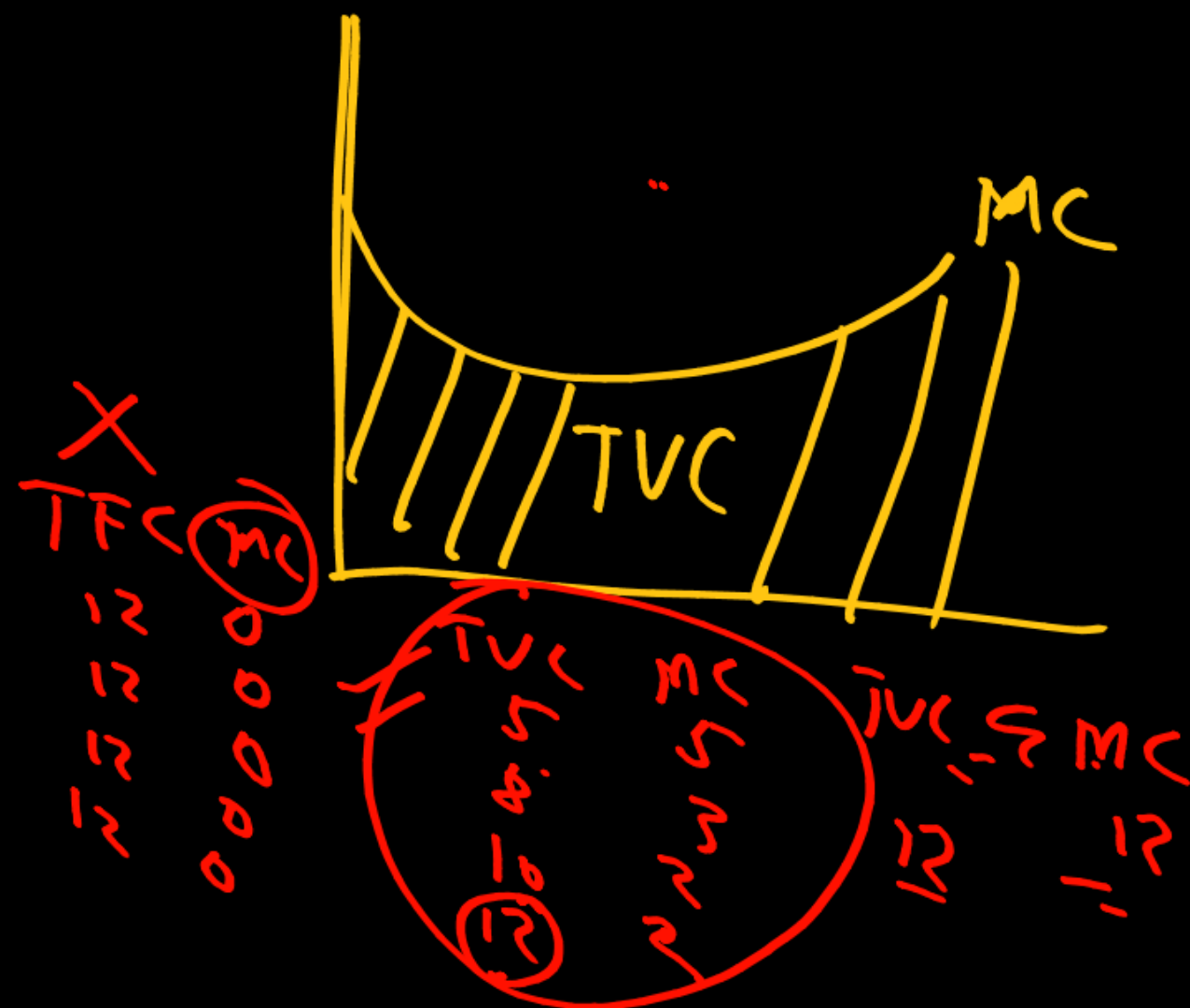
Some Important Observations

- Zero and Negative MR
- $TR = \sum MR$, but $TC \neq \sum MC$

$$TC \neq \sum MC$$



$$TVC = \sum MC$$



Multiple Choice Questions (MCQs)

[Q] Which of the following conditions are true about imperfect competition?

- (a) $AR = MR$ ✓ ✓ → Perfect Competition
- (b) $AR < MR$
- (c) $AR > MR$ ✓
- (d) None of these

Ans. (c) $AR > MR$



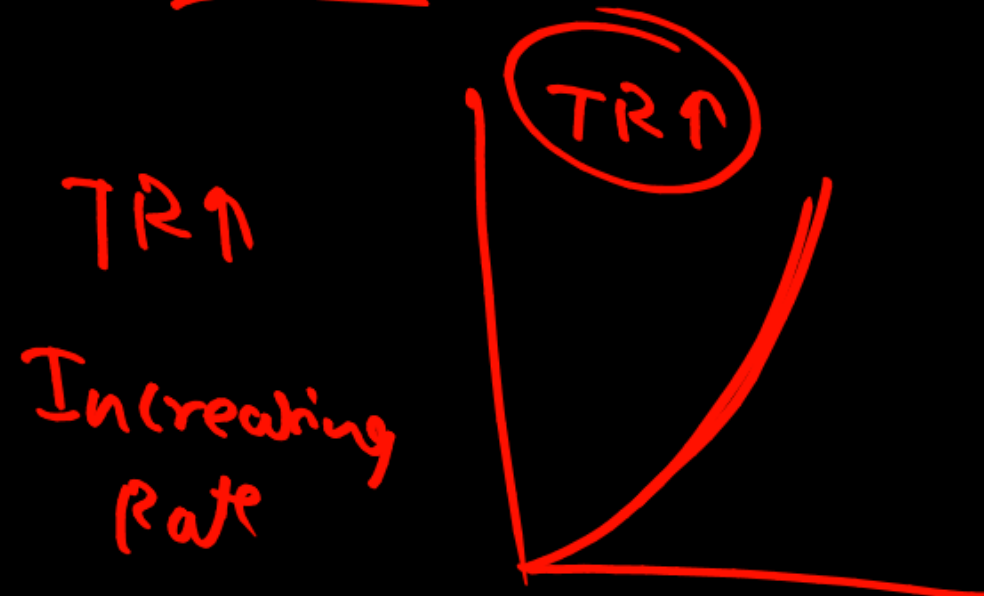
Questions

[Q] What change in total revenue will result in case of

- (i) decrease in marginal revenue?
- (ii) increase in marginal revenue?



- Ans. (i) An increase in total revenue at a diminishing rate or a decrease in total revenue will result from decrease in marginal revenue. In other words, total revenue increases at a decreasing rate or even falls, when marginal revenue falls.
- (ii) An increase in total revenue at an increasing rate will result in an increase in marginal revenue. In other words, marginal revenue increases, when total revenue increases at an increasing rate.



Questions

$P \downarrow, Q \uparrow$



[Q] "Average revenue curve represents law of demand." Comment.

Ans. Average revenue is determined by dividing total revenue by the quantity sold which indicates price of the commodity. Hence, average revenue curve shows the relationship between price of a commodity and quantity demanded.

It is downward sloping curve because to increase its sales, firms have to lower their prices. So, it possesses all the characteristics of the demand curve. Therefore, we can say that average revenue curve represents law of demand.

$$\underline{\text{Price}} = \underline{\text{AR}}$$

$\text{AR} \downarrow, Q \uparrow$

$\underline{\underline{P \downarrow, Q \uparrow}}$

$P \uparrow, Q \downarrow$

$P \downarrow, Q \uparrow$



Question

Calculate Average Revenue (AR) and Marginal Revenue (MR):

Units sold	1	2	3	4	5	6	7
TR (₹)	20	36	48	56	60	60	56

Solution

Unit Sold (Q)	TR (₹)	AR (₹) $TR \div Q = AR$	MR (₹) $TR_n - TR_{n-1} = MR_n$
1	20	20	20
2	36	18	16
3	48	16	12
4	56	14	8
5	60	12	4
6	60	10	0
7	56	8	-4

Question

Complete the following table-

Price (₹)	12	10	8	6
Output (Units)	1	2	3	4
TR (₹)	–	–	–	–
MR (₹)	–	–	–	–

H.W

Solution

Price (P) (₹)	Output (Units) (Q)	TR (₹) $P \times Q = TR$	MR (₹) $TR_n - TR_{n-1} = MR_n$

[illegible]

DOUBT



**THANK
YOU**

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KEEP LEARNING

